LINGVESSA

AI Integration Specification

AI for Learning + AI for Administration / version 1.4

Provider-agnostic, free/local-first architecture for optional AI capabilities without making the core course dependent on AI.

Status: OWNER-LOCKED IMPLEMENTATION BASELINE

Owner lock: 10 September 2026 | Document version date: 09.09.2026 | Replaces AI Integration Specification v1.3

PRODUCT	LINGVESSA
AI MODES	MANUAL / AI ASSIST / AI AUTO
COST DEFAULT	FREE-ONLY; PAID BUDGET = 0
CORE GUARANTEE	LESSON + REVIEW + PROGRESS WORK WITHOUT AI

Паспорт документа

Поле	Значення
Ідентифікатор	LINGVESSA-AI-SPEC
Версія / дата	1.4 / 09.09.2026
Статус	OWNER-LOCKED IMPLEMENTATION BASELINE; owner lock 10.09.2026; P0 authorized.
Governing baseline	Formal Specification v1.7; Technical Specification v1.7; Technical Design v1.7; Curriculum Blueprint A1 v1.6; Reference Module Design v1.6; AI Integration Specification v1.4; Implementation Plan v1.4 (execution order only).
Продукт	Lingvessa; English + Čeština from launch; uk-UA interface/explanations; en-GB default, en-US selectable, cs-CZ.
Мета	Визначити AI Layer для learner-facing та admin-facing функцій, provider routing, privacy, cost control, permissions, testing, observability і graceful degradation.
Ключова межа	AI є optional capability layer. Core curriculum, Lesson Engine, deterministic evaluation, Review, progress, mastery і published media не залежать від доступності LLM.
Precedence	Цей документ доповнює governing baseline. Для AI-specific behavior він має пріоритет над старими формулюваннями, де AI conversation/writing feedback були лише deferred, але не змінює core learning invariants без окремого spec update.

Нормативні слова: MUST - release-blocking requirement; SHOULD - default, відхилення потребує ADR; MAY - сумісне розширення. AI output ніколи не стає source of truth лише тому, що його створила модель.

Що цей документ фіксує вперше

Область	Рішення
AI scope	AI for Learning + AI for Administration у спільному provider-agnostic AI Layer.
Cost	На старті paid inference вимкнено; monthly paid budget = 0. Free/local providers можуть використовуватися лише в межах policy і quota.
Admin modes	Manual, AI Assist, AI Auto з per-capability permissions. AI Auto не має права publish/delete/role/billing у v1.
Learning AI	Conversation/role-play, personalized explanation, adaptive hints, writing feedback, extra practice generation, error-focused remediation - optional and feature-flagged.
Grounding	Learner AI спирається на approved curriculum, knowledge graph, language/variant rules and learner-safe context; model memory is not curricular authority.
Mastery	AI feedback is advisory by default. Mastery updates require validated evaluator/rubric or deterministic activity contract; 6 skill projections + item-level facets remain governing model.
Privacy	Sensitive learner data is routed only to approved providers/data policies; free-tier availability is not a reason to send personal or raw voice data.

Область	Рішення
Media	AI may orchestrate image/audio generation in admin pipeline, but course media remains pre-generated, QA-approved and bound by asset_id + asset_version.

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1. Призначення, scope та незмінні принципи

Lingvessa додає AI як керований capability layer поверх уже визначеної content-driven навчальної системи. AI має підсилювати пояснення, практику, персоналізацію та адміністрування, але не замінювати curriculum, deterministic state machine, published content governance або learner evidence model.

1.1. У межах AI Integration v1.4

- AI text generation and structured reasoning for learner-facing and admin-facing functions.
- Provider adapters for cloud free tiers, local models and optional paid providers later.
- AI conversation and role-play in text; voice mode through existing STT/TTS/speech adapters when enabled.
- Personalized explanation, adaptive hinting, writing feedback and additional practice generation.
- Admin-assisted content drafting, QA suggestions, metadata completion, media prompt/script generation, analytics summaries and controlled low-risk automation.
- AI settings, provider routing, feature flags, quotas, audit, privacy classification, safety checks and evals.

1.2. Поза межами цього baseline

- Autonomous AI publication of curriculum without human editorial gates.
- AI deletion of user accounts, role changes, billing changes or irreversible production operations.
- Live image/audio generation as a dependency of ordinary lesson completion.
- Using model output as official CEFR certification or as unvalidated high-stakes assessment.
- Permanent storage of hidden model reasoning/chain-of-thought. Only final outputs, structured traces and policy metadata needed for operation may be stored.
- Assumption that any specific free provider or model will remain free indefinitely.

Core product invariant: if every AI provider is disabled, rate-limited or offline, the user must still be able to log in, open published lessons, complete deterministic exercises, receive normal feedback, save progress, use Review, collections, audio assets and resume learning.

ID	Вимога	Пріоритет	Критерій перевірки
AI-GOV-001	AI must be optional for core learner path and cannot be a hidden runtime dependency of published lessons.	MUST	Provider-offline E2E completes both English and Czech reference modules excluding explicitly optional AI activities.
AI-GOV-002	AI output is non-authoritative until validated by a defined contract, schema, evaluator or human review gate.	MUST	No AI endpoint can directly set published=true, correct=true, mastery estimate or role.
AI-GOV-003	AI-specific behavior is versioned and traceable to provider/model/prompt/policy versions.	MUST	Audit/support can reproduce configuration of a representative AI result.
AI-GOV-004	Provider choice remains implementation detail behind internal ports.	MUST	Domain and content schemas contain no provider SDK types or canonical provider URLs.

2. AI capability model

Domain	Capability	Default v1	Learning/operational authority
Learning	AI explanation / "Пояснити інакше"	OFF until learner opt-in; then ON only when capability/provider policy is approved	Advisory; grounded in approved curriculum.
Learning	Adaptive hints	ON selectively	Advisory; cannot reveal more than activity policy permits.
Learning	Extra practice generator	ON selectively	Practice-only unless generated from validated deterministic template.
Learning	Writing feedback	Feature flag	Advisory rubric feedback; mastery only through validated assessment contract.
Learning	AI conversation / role-play	Feature flag	Practice/interactions; speaking/writing evidence rules apply separately.
Learning	Voice AI role-play	Later flag	STT -> AI -> TTS pipeline; no realtime provider lock-in required.
Admin	Draft lessons/exercises/explanations	AI Assist default	Draft only; publish gates unchanged.
Admin	QA suggestions / anomaly analysis	AI Assist default	Creates issues/proposals, not authoritative fixes.
Admin	Media prompt/script generation	AI Assist / Auto low-risk	Can create jobs/drafts; media QA and rights gates required.
Admin	Auto-fix low-risk draft metadata	AI Auto allowlist	Draft only; diff and audit required.
Admin	Publish / rollback / user delete / role	Manual only	AI prohibited from direct execution in v1.

2.1. Capability risk classes

Class	Examples	Control
A - low risk	summarize analytics, draft tags, generate internal prompt, suggest alt text	May run AI Auto on draft/non-sensitive data with audit.
B - bounded learner assistance	explanation, hint, extra practice, conversation	Feature flag, grounding, output validation, rate limit, learner transparency.
C - learning judgment	writing score, pronunciation interpretation, placement recommendation	Validated rubric/evaluator, confidence/fallback, no single-model authority.
D - privileged/destructive	publish, delete, role change, billing, legal consent	Manual human action only in v1; AI may summarize/propose.

3. Core-without-AI invariant i graceful degradation

AI availability is best-effort. Every AI entry point defines a non-AI fallback. A 429, provider outage, model removal, quota exhaustion, invalid JSON, safety refusal or timeout cannot corrupt lesson state or create fake mastery evidence.

AI feature	Fallback
Personalized explanation	Show curated explanation, examples and usage notes from published content.
AI hint	Use deterministic hint/reveal policy already stored in activity.
Extra practice	Use scheduled Review items and curated remediation templates.

AI feature	Fallback
AI conversation	Offer scripted dialogue/role-play or text practice from module.
Writing feedback	Save response; show deterministic checks where available; mark AI feedback unavailable.
Admin content draft	Editor continues manual structured authoring.
AI QA	Normal schema/language/pedagogical/media QA gates still run.
AI media orchestration	Media requirement remains queued for manual/alternative provider handling.

ID	Вимога	Пріоритет	Критерій перевірки
AI-FALL-001	Every learner AI feature has explicit fallback UX and safe status.	MUST	Fault matrix for timeout/429/5xx/provider disabled produces no dead end.
AI-FALL-002	AI failure never rolls back an already accepted deterministic learning transaction.	MUST	Injected AI failure after attempt commit leaves progress unchanged and durable.
AI-FALL-003	Fallback routing must respect the same data-class/privacy policy; a cheaper/free provider cannot bypass it.	MUST	Sensitive request is rejected rather than routed to unapproved provider.

4. AI Layer architecture

AI інтегрується як окремий application module з internal ports. Learner UI і Admin/CMS звертаються до Lingvessa API, а не безпосередньо до provider SDK.

Web/PWA / Admin -> Application API -> AI Orchestrator -> Policy + Grounding + Router -> Provider Adapter(s) -> Model
-> Validator / Safety / Usage Meter -> Result + Audit

Component	Responsibility	Must not do
AI Orchestrator	Resolve capability, data class, prompt template, grounding, routing, output schema and fallback.	Must not bypass policy or directly mutate published content/mastery.
AI Policy Engine	Permissions, sensitivity, free-only/paid budget, provider allowlist, model capability rules.	Must not depend on client-provided role/price authority.
Grounding Builder	Fetch approved curriculum/knowledge/content context and minimal learner state.	Must not send unrelated profile/history.
Provider Router	Choose approved provider/model by capability, privacy, quota, latency and cost.	Must not route to provider with incompatible data policy.
Provider Adapter	Translate internal request/result to provider API.	No SDK types leak into domain schemas.
Output Validator	JSON/schema, language/variety, safety, citation/grounding, hallucination heuristics.	Must not silently coerce invalid language answer into approved content.
Usage Meter	Tokens/requests/audio seconds/estimated cost and hard budgets.	Must not permit automatic paid overage when paid disabled.
AI Audit	Record actor, purpose, provider/model/prompt/policy versions, result class and action proposal.	No secrets, raw chain-of-thought, raw mic bytes.

4.1. Internal request lifecycle

1. Classify capability and actor.
2. Resolve target language/variety and data sensitivity.
3. Check feature flag, permission and budget.
4. Build minimal grounded context.
5. Select provider/model through policy-aware router.
6. Execute with timeout, retry/circuit-break rules.
7. Validate structured output and safety.
8. Return learner/admin result; persist only approved metadata/content according to retention policy.
9. Emit usage/audit/quality events.

5. Provider abstraction, routing i free/local-first policy

The system MUST not assume one permanent AI vendor. Provider selection is configuration, not product identity. Start policy is free/local-first with hard protection against accidental paid usage.

5.1. Provider classes

Class	Use	Advantages	Constraints
Local model	Admin drafts, QA, selected learner text features	No per-request cloud API fee; maximum data control.	Hardware-dependent quality/latency; model management; not automatically scalable.
Free cloud API	Development, low-volume learner/admin requests	No inference charge within provider limits.	Rate limits, terms/data policy may change; no production SLA.
Aggregator/router free	Experimentation/fallback	Many models through one API.	Underlying provider/data policy can vary; low daily quotas may apply.
Paid cloud provider	Optional future production	Higher limits, model quality, support.	Disabled by default; explicit budget and approval required.

5.2. Routing decision order

10. Capability compatibility (text, structured output, tools, vision, speech if relevant).
11. Data-class/privacy eligibility.
12. Model quality and language suitability for Ukrainian + English/Czech task.
13. Free-only / paid policy and remaining quota.
14. Latency and provider health.
15. Fallback priority and circuit-break state.

ID	Вимога	Пріоритет	Критерій перевірки
AI-PROV-001	Each provider is configured through adapter + provider policy profile.	MUST	Swap adapter without changing learner/content schema.
AI-PROV-002	The router evaluates data policy before price/latency.	MUST	Sensitive fixture never reaches cheaper but unapproved provider.
AI-PROV-003	Free-tier/provider terms are treated as volatile configuration and revalidated before production use.	MUST	Provider registry records verified_at, terms/data-policy refs and owner.
AI-PROV-004	Model deprecation/unavailability produces fallback or AI-unavailable, never core failure.	MUST	Disable primary model during E2E and complete non-AI flow.

6. Cost, quotas i protection from accidental billing

Start default: paid_inference_enabled = false; monthly_paid_budget = 0; free_only = true. Exhausting a free quota disables/falls back the AI capability. It MUST NOT silently switch to a paid model.

Control	Default	Behavior
paid_inference_enabled	false	No provider/model with non-zero expected cost is eligible.
monthly_paid_budget	0	Hard server-side budget; not a UI-only warning.
free_only	true	Router only uses provider/model profiles tagged currently_free and approved.
daily_request_cap	Per capability/provider	Protects free quotas and abuse.
per_user_ai_cap	Conservative	Prevents one learner consuming shared free quota.
admin_bulk_dry_run	true	Shows estimated requests/tokens/assets before execution.
automatic_fallback	Approved free/local only	No paid fallback unless owner explicitly changes policy.
budget_alerts	80/95/100%	At 100%, cost-bearing AI is blocked.

ID	Вимога	Пріоритет	Критерій перевірки
AI-COST-001	No paid API charge can occur with paid_inference_enabled=false.	MUST	Integration test with paid candidate selected returns POLICY_BLOCKED before network call.
AI-COST-002	All AI calls emit usage metadata and estimated/actual cost where provider reports it.	MUST	Usage dashboard reconciles provider response headers/usage fields for sampled calls.
AI-COST-003	Bulk AI jobs require dry-run and bounded batch size.	MUST	Admin can preview quota impact and abort before provider call.

7. AI for Learning

7.1. Personalized explanation

Entry point examples: "Пояснити інакше", "Чому тут так?", "Дай ще один приклад". AI receives the current knowledge_item, approved explanation/usage notes, target language/variety, CEFR scope and minimal mistake context. It explains in uk-UA by default and must stay within the learner level unless explicitly asked to extend.

ID	Вимога	Пріоритет	Критерій перевірки
AI-LRN-001	AI explanation must be grounded in approved knowledge/content IDs and return grounding_refs.	MUST	Response without valid refs is rejected or downgraded to generic non-authoritative assistance.
AI-LRN-002	AI must not contradict accepted variants, English variety or Czech morphology rules in published content.	MUST	Golden bilingual eval suite passes approved rule cases.

7.2. Adaptive hints

AI may choose or phrase a hint only within the activity hint_policy. It MUST NOT expose the full answer before reveal is allowed. Hint usage remains assistance evidence and affects scheduling according to the existing deterministic rules.

7.3. Extra practice generator

AI can generate ephemeral practice from specific knowledge_item IDs and weak facets. Each generated activity carries provenance, model/prompt version, target operation and response contract. Practice generated from free-form AI is practice-only by default. If an item uses an approved template with deterministic evaluator and passes schema/answer validation, it MAY create normal evidence at a capped weight.

Generation mode	Mastery effect
Free-form AI question	No direct mastery; practice feedback only.
Template-constrained AI surface + deterministic evaluator	May create bounded evidence after validation.
Admin-generated + reviewed + published item	Normal course evidence under published content version.

7.4. Writing feedback

AI feedback should separate: task completion, meaning, grammar/form, vocabulary/chunks, register, spelling/punctuation and suggested rewrite. The learner original is preserved according to privacy policy. A polished rewrite must be clearly marked as a suggestion, not silently substituted for the learner response.

- For A1/A2, use short, concrete feedback and one or two highest-value corrections.
- For B1-C1, allow richer rubric feedback, register and cohesion comments.
- AI writing score is not authoritative mastery unless a calibrated rubric/evaluator version is approved.

7.5. AI conversation / role-play

Conversation is scenario-bounded, not an unrestricted chatbot by default. It receives CEFR target, communicative goal, permitted knowledge range, character role, difficulty, register and correction mode. It should recycle known material and introduce limited supported novelty.

Mode	Behavior
Guided text role-play	Short turns, optional hints, scenario goal, end summary.
Free text conversation	Broader but still target-language/level aware; learner may choose topic.
Voice role-play	STT -> AI turn -> TTS; speaker/avatar synchronization; speech evidence handled separately.
Correction mode	Immediate, delayed-summary, or minimal-interruption; user setting.

AI conversation does not replace the curated core course. It is an optional practice surface linked to the same knowledge graph and learner profile.

7.6. AI and pronunciation/speaking

AI Integration does not override the Lingvessa speech contract from Technical Design v1.7. For usable speech evidence, the learner-facing UX may show correct / try again. Low confidence, mic denial or provider failure becomes speech_unavailable, neutral and non-blocking. Raw scores need not be shown. AI may provide a short explanation of a detected target-feature issue only when the speech evaluator returns usable structured evidence.

8. Grounding, learner context i knowledge safety

The model must not rely on its general memory as curricular truth when Lingvessa already has approved content. The Grounding Builder assembles only the context needed for the request.

Context source	Examples	Rule
Published curriculum	lesson outcome, explanation, accepted variants, usage notes	Highest pedagogical authority for learner AI.
Knowledge graph	prerequisites, contrasts, chunks, language-specific facets	Use stable IDs and exact versions.
Learner model	weak skill/facet, recent mistakes, selected topic	Send minimal derived state, not full history.
Language profile	target language, en-GB/en-US, CEFR path	Mandatory scope.
Conversation state	recent AI turns or compact summary	Bounded retention and size.
External web	Not required for ordinary language teaching	Disabled by default unless a separate feature explicitly needs fresh facts.

ID	Вимога	Пріоритет	Критерій перевірки
AI-GRD-001	Grounded learner request includes exact content/knowledge versions where relevant.	MUST	Audit result identifies source IDs/versions.
AI-GRD-002	Learner context is minimized to purpose; no broad export of account/history.	MUST	Privacy test inspects outbound payload allowlist.
AI-GRD-003	If grounding is unavailable or contradictory, AI returns uncertainty/fallback instead of inventing a rule.	MUST	Broken grounding fixture produces safe refusal or curated fallback.

9. AI evidence, mastery i assessment rules

Lingvessa retains the two-level mastery model fixed in Technical Design v1.7: six global skill projections (Vocabulary, Grammar, Listening, Speaking, Writing, Pronunciation) plus the canonical knowledge-item facet registry (e.g. recognition, meaning, recall_l2_to_l1, recall_l1_to_l2, production, spelling, orthography, listening_recognition/listening_gist/listening_detail, pronunciation_target, intelligibility, context_use, collocation, grammar_form, word_order, morphology, register, variant_awareness, interaction_repair, interaction_goal). Every evidence target must use an allowed skill-facet pair.

AI signal	Default evidence effect
AI explanation viewed	No positive mastery by itself.
AI hint used	Assistance metadata; does not equal independent retrieval.
AI-generated free-form practice answer	Practice-only unless evaluator contract is validated.
Template-constrained item evaluated deterministically	Bounded evidence to exact skill/facet.
AI writing feedback	Advisory; no direct mastery unless calibrated rubric accepted.
AI conversation text turn	May count as production/interaction only through approved rubric/evaluator; not automatic.
Voice turn	Speaking/Pronunciation evidence only if speech subsystem produces usable evidence; low confidence = none.

ID	Вимога	Пріоритет	Критерій перевірки
AI-MAST-001	LLM self-reported confidence never directly changes mastery.	MUST	Attempt/evidence API rejects AI confidence as authority.
AI-MAST-002	AI-generated evidence targets exact global skill + item facet where applicable.	MUST	Evidence fixtures distinguish meaning vs spelling vs production.
AI-MAST-003	High-stakes level readiness cannot depend on one uncalibrated AI scorer.	MUST	Level Test config has deterministic/rubric evidence and calibration record.

10. AI for Administration

Admin AI reduces repetitive work while preserving versioning, review and audit. It operates on drafts, reports and jobs; human responsibility remains for publication and privileged actions.

Capability	AI output	Can AI Auto apply?
Create lesson draft	Structured lesson/activity JSON linked to outcomes/knowledge	Yes, to draft only, if schema valid.
Generate distractors	Candidate alternatives + rationale	Draft only; ambiguity validator + linguistic review required.
Rewrite explanation	Candidate uk-UA explanation at CEFR scope	Draft only.
Generate examples/dialogues	Candidate content with language/variant metadata	Draft only.
Generate media requirements/prompts/scripts	Structured job specification	Yes, low-risk job creation if quota/rights policy satisfied.
Detect duplicates/gaps	Issue list linked to IDs	Yes, issue creation only.
Language QA suggestions	Potential grammar/register/diacritic problems	Suggestion only; no silent publish edit.
Analyze item errors/drop-off	Summary and candidate causes	Yes, report generation.
Create release summary	Diff/risks/checklist summary	Yes, report generation.
Publish/rollback	Proposal/checklist only	No direct execution in v1.
Delete account/change role/billing	No execution	Never in AI Auto v1.

10.1. AI-generated content workflow

Course map -> AI draft -> schema validation -> automated language/content checks -> linguistic review -> pedagogical review -> media generation/QA -> preview -> human approval -> publish

AI may accelerate any draft or QA-preparation step, but published content remains an immutable reviewed snapshot. Every AI-authored or AI-edited field stores generation provenance or diff metadata sufficient for review.

11. Manual / AI Assist / AI Auto permissions

Mode	Behavior	Default
Manual	AI disabled for admin mutation. Optional "Ask AI" may be disabled entirely. Human edits and triggers jobs.	Available always.
AI Assist	AI creates proposals/drafts/diffs; human presses Apply/Approve/Generate.	Recommended default.

Mode	Behavior	Default
AI Auto	Only allowlisted low-risk actions run automatically within quota and draft scope.	Off by default; opt-in per capability.

11.1. Permission matrix

Action	Manual	AI Assist	AI Auto
Suggest lesson/exercise	Human	AI proposal	May create draft
Edit published content	Human via new version	AI proposal only	Prohibited
Create media generation job	Human	Confirm AI proposal	Allowed if low-risk + quota
Approve linguistic/pedagogical gate	Human reviewer	AI may summarize issues	Prohibited
Publish / rollback	Human admin	AI may prepare preflight summary	Prohibited
Delete/disable account	Human authorized	AI may explain scope	Prohibited
Change roles/MFA/security	Human authorized	AI may recommend	Prohibited
Change AI provider/budget	Human authorized	AI may recommend	Prohibited
Create issue from anomaly	Human	AI proposal	Allowed

ID	Вимога	Пріоритет	Критерій перевірки
AI-ADM-001	AI Auto is allowlist-based, not blanket autonomous access.	MUST	Unknown action denied by policy engine.
AI-ADM-002	AI service identity lacks publish, role, account-delete and billing permissions.	MUST	Authorization matrix tests fail direct calls.
AI-ADM-003	Every applied AI change to draft content has diff, actor/mode, model/prompt version and timestamp.	MUST	Admin audit opens complete change trail.
AI-ADM-004	Mode can be changed globally and per capability without redeploy.	SHOULD	Admin setting takes effect on next job/request.

12. AI content and media generation workflow

The AI Layer may generate or orchestrate images/audio only through the existing Media Pipeline. Learners are never asked to create or source required media. Live generation remains outside ordinary lesson runtime.

Stage	AI role	Blocking gate
Media requirement	Infer needed asset from activity semantics	Requirement must be explicit and accessibility role known.
Prompt/script draft	Generate original scene/voice script specification	No proprietary source copying; checksum/version.
Provider selection	Choose approved image/TTS/local generator	Rights/data/cost policy and quota.
Generation	Submit deterministic job	Idempotent job key; provider metadata captured.
Automated QA	Technical checks, duplicate/logo/text detection where available	Required checks pass.
Human/media QA	Semantic correctness, no answer leakage, cultural and voice consistency	Approval required for publish.
Persist	Create immutable asset_version	Concrete storage key/checksum/provenance/rights.
Bind	Attach asset_id + asset_version to content version	No wildcard/provider URL in published manifest.

All required lesson images may be generated by Lingvessa tooling/assistant in advance and stored in the platform media structure. The product owner does not need to manually search for photos.

13. Privacy, minors, data routing and retention

Free inference is not automatically privacy-appropriate. Provider eligibility is determined by data policy, not price. Learner-facing AI may process personal text, conversation context or voice-derived transcripts, so outbound data is classified before routing.

Data class	Examples	Default routing
Public/product	published lesson text, synthetic examples, media prompts	Approved local or free cloud provider.
Pseudonymous learning	knowledge IDs, error categories, CEFR, derived weak facets	Approved provider; no direct identifiers.
Personal learner text	free writing, AI conversation turns	Only provider profile approved for learner content; retention minimized.
Voice/raw audio	microphone bytes	Speech-specific approved processor; transient by default; not sent to generic LLM unless explicitly designed and approved.
Sensitive/regulated	health, legal, account/security data accidentally entered	Minimize/block or local/approved high-privacy path; no casual free-tier routing.
Admin secrets	API keys, tokens, private credentials	Never sent to model.

13.1. Learner AI retention

- Conversation history is optional and has a bounded retention setting; user can clear it.
- For personalization, prefer compact derived summaries (e.g. weak facets) over storing every raw AI turn indefinitely.
- Raw microphone audio remains governed by speech policy and is transient by default.
- Provider logs/training/data-use terms are recorded in provider registry and reviewed before enabling a learner-facing route.
- If minors may use the service, AI features, consent, data processing and age flow require jurisdiction-specific review before public launch.

ID	Вимога	Пріоритет	Критерій перевірки
AI-PRIV-001	Outbound AI payload is built from an allowlist by capability/data class.	MUST	Privacy fixture detects no unrelated profile, tokens or raw audio.
AI-PRIV-002	Provider profile records retention/training/data-use eligibility and last verification date.	MUST	Routing blocks provider with unknown policy for learner personal data.
AI-PRIV-003	AI conversation deletion/retention propagates to Lingvessa stores and supported provider controls according to policy.	MUST	Deletion drill reports completed/failed downstream steps.

14. Security, prompt injection and abuse controls

AI introduces a new untrusted-input/output boundary. Learner content, external documents, model output and provider callbacks are treated as untrusted. System instructions, secrets and privileged tools are never delegated to the model without server-side authorization.

Threat	Control
Prompt injection from learner text	Separate system policy from learner content; no privileged tool execution from raw model instructions.
Prompt injection from content/imported text	Only approved internal context; content is data, not executable instruction.
Tool abuse	Server validates tool name, arguments, actor permissions and object scope; deny by default.
Secret exfiltration	Secrets never included in prompt; egress allowlist and log redaction.
Model hallucination	Grounding refs, schema validation, language QA/evals, uncertainty fallback.
Unsafe/inappropriate role-play	Age-aware content policy, scenario restrictions, moderation where needed.
Cost abuse	Rate limits, per-user caps, free-only/paid budget hard gate.
Provider callback spoofing	Signature/timestamp/job match/replay window as in existing job contract.
Data poisoning via AI Auto	Draft-only writes, version diff, validator gates, human publish approval.

ID	Вимога	Пріоритет	Критерій перевірки
AI-SEC-001	AI cannot execute privileged operations solely because model output requests them.	MUST	Red-team prompt cannot publish/change role/delete.
AI-SEC-002	Tool calls use server-side allowlist + schema + RBAC/object authorization.	MUST	Invalid/unauthorized tool call rejected and audited.
AI-SEC-003	Prompts and outputs are excluded from error telemetry unless explicitly allowlisted/redacted.	MUST	Canary secret/raw learner text absent from telemetry sink.

15. Data model, APIs and jobs

15.1. New domain entities

Entity	Purpose / key fields
AIProviderProfile	provider_id, adapter_type, status, capabilities, data_policy_class, free_status, verified_at, regions, retention/training flags, quota metadata.
AIModelProfile	model_id, provider_id, capability tags, languages, context, structured-output/tool support, cost/free metadata, status/version.
AIPolicy	capability, actor role, data class, allowed providers/models, free_only, paid_budget, retention, fallback policy, version.
AIPromptTemplate	purpose, language/variant scope, template version, system rules, output_schema_code, grounding policy, status.
AIRun (logical request record)	request_id, actor class, capability, language profile ref if needed, sensitivity, provider/model/prompt/policy versions, timing, result class, usage/cost; no secret/CoT.
AISession	optional learner conversation session, language, scenario, state/summary, retention expiry, consent/policy refs.
AIActionProposal	admin proposed action, target draft/version, diff, risk class, status, approval actor/time.
AIEvalCase	fixture input, expected constraints, language, capability, version, pass criteria.

Entity	Purpose / key fields
AIUsageBucket	provider/model/capability/user cohort/day usage, quota and estimated cost.

15.2. API surface

Method/path	Purpose	Critical contract
POST /api/v1/ai/explain	Grounded learner explanation	Auth/profile ownership; knowledge/content refs; no mastery mutation.
POST /api/v1/ai/hint	Bounded hint	Must respect activity hint/reveal policy.
POST /api/v1/ai/practice	Generate ephemeral practice	Schema validation; practice/evidence policy explicit.
POST /api/v1/ai/writing-feedback	Rubric feedback	Retention/data policy; result advisory unless evaluator approved.
POST /api/v1/ai/conversations	Start scenario	Language/CEFR/goal; quota; AI label; retention state.
POST /api/v1/ai/conversations/{session_id}/messages	Continue turn	Owner/session; bounded context; safe output.
GET/PATCH /api/v1/admin/ai/settings	AI modes/providers/quotas	Admin/MFA; audited; no secrets returned.
POST /api/v1/admin/ai/proposals	Create draft/admin proposal	Role/target/version; no publish.
POST /api/v1/admin/ai/jobs	Bulk generation/QA job	Dry-run, deterministic key, quota/cost, bounded retry.

15.3. Job classes

Long-running AI generation uses workers/queues with deterministic keys, timeout, bounded retry and dead-letter state. AI jobs do not run inside the core attempt database transaction.

16. Prompt/template/version governance

Prompts are production artifacts. They are versioned, reviewed and testable. An unnoticed prompt change can alter learning behavior as much as a code change.

Artifact	Required metadata
Prompt template	id, version, purpose, actor type, language/variety scope, system rules, expected output schema, grounding policy, safety policy, owner, status.
Few-shot examples	versioned examples, provenance, language QA, no private learner data.
Output schema	JSON Schema version, strictness, migration path.
Model profile	provider/model version, capability, quality baseline, retirement status.
Eval suite	golden cases, adversarial cases, language-specific cases, pass threshold/version.

ID	Вимога	Пріоритет	Критерій перевірки
AI-PRM-001	Prompt/template change triggers relevant eval suite before rollout.	MUST	CI/staging report links prompt version to pass results.
AI-PRM-002	Prompt includes no secrets or unversioned private examples.	MUST	Secret/privacy lint passes.

ID	Вимога	Пріоритет	Критерій перевірки
AI-PRM-003	Model switch is a controlled config release with eval evidence for critical capabilities.	MUST	Router cannot promote untested model to learner default.

17. UX, accessibility and transparency

17.1. Learner UX

- AI entry points are clearly optional and labeled, e.g. "AI-пояснення" or "Розмовна практика з AI".
- The learner sees a concise unavailable state if AI is offline/quota exhausted, plus a non-AI alternative.
- AI-generated correction distinguishes learner original, correction and explanation.
- Conversation offers correction intensity/mode and text/voice availability according to capability.
- AI never hides the ordinary Reveal/Retry/Continue rules of scored core activities.
- Accessibility: keyboard, screen-reader status, reduced motion, captions/transcripts and non-mic alternatives remain available.

17.2. Admin UX

- Global AI enabled toggle plus separate Learning AI and Admin AI toggles.
- Mode selector Manual / AI Assist / AI Auto.
- Per-capability switches and risk labels.
- Provider/model priority, free-only setting, quota and budget dashboard.
- Data-policy eligibility visible for learner vs admin tasks.
- Proposal diff before Apply; audit link after action.
- One-click emergency disable for provider and for all AI.

EU launch note: AI interaction should be transparently identified to users, and Article 50 AI Act obligations must be evaluated per flow/provider/deployer role. The platform should not rely on users guessing that a conversational partner is artificial.

18. Observability, quality metrics and incident handling

Signal	Examples	Rule
Usage	requests, tokens, audio seconds, quota remaining, free/paid classification	Bounded labels; cost protection.
Reliability	latency, 429, 5xx, timeout, fallback rate, invalid-schema rate	Per provider/model/capability.
Learning quality	helpfulness feedback, correction acceptance, repeated error after AI help, conversation completion	No raw learner content in metrics.
Content quality	AI draft rejection reason, linguistic/pedagogical edit rate, hallucination/ambiguity reports	Compare model/prompt versions.
Safety/privacy	policy blocks, sensitive-route rejects, moderation/abuse events	Restricted access and retention.
Admin autonomy	AI Auto actions, reversions, proposal acceptance rate	Every automatic action traceable.

ID	Вимога	Пріоритет	Критерій перевірки
AI-OBS-001	Every AI call has correlation_id and provider/model/prompt/policy versions.	MUST	Support can trace request without raw secret content.
AI-OBS-002	Cost/quota telemetry distinguishes free vs potentially billable calls.	MUST	Dashboard and alert show zero paid calls under start policy.
AI-OBS-003	Provider or AI subsystem can be disabled by feature flag without redeploy.	MUST	Incident drill disables AI and core lesson remains usable.

19. Testing, evals and release gates

Layer	Scope	Gate
Unit	policy/routing/budget/schema/fallback decisions	Every code change.
Contract	provider adapters, structured output, usage/cost mapping	Adapter/model integration.
Bilingual evals	uk-UA explanation + en-GB/en-US + cs-CZ grammar/usage	Prompt/model change.
Learning evals	helpfulness, level appropriateness, no answer leakage, rubric consistency	Learner AI release.
Admin evals	schema-valid drafts, ambiguity, originality, rights metadata	Admin AI release.
Security red-team	prompt injection, tool abuse, secret exfiltration, role escalation, data routing	Public release.
Privacy	payload allowlists, retention, provider policy, deletion	Learner AI release.
Cost/fault	quota exhaustion, 429, timeout, provider removal, paid disabled	Release candidate.
Accessibility	AI states, conversation, proposal UI, keyboard/screen reader	UI release.

19.1. Minimum bilingual golden cases

- English en-GB explanation and en-US variant note do not get mixed.
- Czech diacritics, ty/vy register, být forms and origin morphology are explained correctly.
- AI does not mark a Czech wrong case as vocabulary-meaning failure.
- AI distinguishes English meaning-correct full form from contraction-form objective.
- AI explanation stays within A1 when requested from A1 reference module.
- No prompt tells the learner to copy Babbel or reproduces proprietary course text.
- Low confidence speech remains speech_unavailable and does not become a language judgment.

20. Implementation sequence

Stage	Technical output	Exit evidence
AI0. Contracts	AI ports, schemas, provider registry, policy engine, prompt registry, usage meter, audit model.	No provider call yet; unit/contract fixtures pass.
AI1. Local/free adapter spike	At least one local and one free cloud text adapter behind same port.	Same structured request passes through both; paid calls blocked.
AI2. Admin AI Assist	Draft/proposal/QA/report functions in CMS.	Diff/audit, schema validation, no direct publish.

Stage	Technical output	Exit evidence
AI3. Learner explanation + hints	Grounded explain/hint endpoints for E-A1-M01 and C-A1-M01.	Bilingual evals + fallback + privacy tests.
AI4. Extra practice	Ephemeral practice generator linked to knowledge/facets.	No false mastery; deterministic-template path tested.
AI5. Writing feedback	Versioned rubric feedback.	A1/A2 cases, retention and advisory scoring tested.
AI6. Conversation text	Bounded role-play linked to module goals.	Scenario quality, rate limits, AI label, fallback.
AI7. Voice conversation	Integrate STT/TTS/speech adapters if useful/cost-feasible.	Mic/privacy/failure tests; core unaffected.
AI8. AI Auto low-risk admin	Allowlisted draft/jobs/issues automation.	No privileged action possible; audit/rollback of draft changes.
AI9. Production hardening	Provider review, quotas, eval dashboards, incident runbook, transparency/legal review.	All acceptance scenarios pass.

Programming priority: build the AI foundation after core contracts are stable enough to expose knowledge/content/version data. Do not delay the non-AI reference implementation waiting for AI.

21. Acceptance scenarios

ID	Вимога	Пріоритет	Критерій перевірки
AIA-01	All AI providers disabled	MUST	English and Czech core lesson/review/progress flows still complete; AI buttons show fallback/unavailable only.
AIA-02	Free quota exhausted	MUST	No paid call occurs; route to approved local/free fallback or return unavailable.
AIA-03	Sensitive learner text with unapproved free provider first in priority	MUST	Router skips/blocks provider; data is not sent.
AIA-04	Grounded explanation asks why Czech form differs	MUST	Response cites exact knowledge/content refs and follows cs-CZ rule; invalid grounding is rejected.
AIA-05	English en-US selected	MUST	AI explanation/role-play respects en-US while shared curriculum semantics remain.
AIA-06	AI extra practice free-form	MUST	Item is marked ephemeral/practice-only; no direct mastery update.
AIA-07	Template-constrained generated item	MUST	Schema/evaluator validation passes; bounded facet evidence may be recorded.
AIA-08	Writing feedback model returns invalid JSON	MUST	Validator retries/fallbacks; learner original saved; no corrupted score/mastery.
AIA-09	Prompt injection asks AI admin to publish and change role	MUST	Tool/policy engine denies; event audited.
AIA-10	AI Auto creates draft exercise and media job	MUST	Only draft/job changes occur; diff/audit/quota present; publish still human.
AIA-11	Provider model removed	MUST	Router fallback works or feature unavailable; core unaffected.

ID	Вимога	Пріоритет	Критерій перевірки
AIA-12	Mic denied/low confidence during AI role-play	MUST	speech_unavailable; text/non-mic route offered; no speaking/pronunciation penalty.
AIA-13	AI conversation cleared by user	MUST	Lingvessa conversation state removed/expired and downstream handling follows policy.
AIA-14	Paid provider configured but paid_inference_enabled=false	MUST	Network call blocked before provider; usage log shows policy block and zero cost.
AIA-15	Prompt/model version changed	MUST	Relevant bilingual/security evals required before learner-default promotion.
AIA-16	AI media prompt generates required lesson image	MUST	Asset enters existing media pipeline; cannot publish until concrete approved asset_version exists.

22. Fixed decisions and ADR backlog

22.1. Fixed decisions carried forward into AI Integration v1.4

- Lingvessa has one AI Layer for both Learning and Administration.
- Core learner path works without AI.
- Provider-agnostic routing; local/free-first; no hidden vendor lock-in.
- Paid AI is disabled by default; budget = 0 until owner explicitly changes it.
- AI Assist is recommended admin default; AI Auto is opt-in and allowlist-based.
- AI cannot directly publish, rollback, delete accounts, change roles or billing in v1.
- Learner AI is grounded in approved curriculum/knowledge and language profile.
- AI output is advisory unless validated by deterministic evaluator/rubric/human gate.
- Two-level mastery model remains authoritative; AI does not collapse skills/facets.
- Speech contract from Technical Design v1.7 remains: usable speech may show correct/try again; unavailable/low-confidence neutral and non-blocking.
- AI-generated images/audio for course content are pre-generated through Media Pipeline, QA-reviewed and versioned before publish.
- No provider free tier is assumed permanent.

22.2. ADRs required later

ADR	Decision	Default / deadline
AI-ADR-001	Initial text provider(s)	Choose after current free-tier + data-policy + quality spike; before AI1.
AI-ADR-002	Local model runtime	Ollama-compatible adapter candidate; before AI1 local spike.
AI-ADR-003	Embedding/RAG implementation	PostgreSQL-compatible/vector option only if retrieval needs measured; before AI3.
AI-ADR-004	Learner conversation retention	Minimal/off by default; decide before AI6 public use.
AI-ADR-005	Writing evidence authority	Advisory until pilot/calibration; before any mastery-bearing AI rubric.
AI-ADR-006	Voice conversation provider topology	STT -> LLM -> TTS as portable default; realtime speech-to-speech only after need/cost evaluation.
AI-ADR-007	Production paid AI budget	Remain 0 until product owner explicitly approves paid operation.

ADR	Decision	Default / deadline
AI-ADR-008	EU AI transparency implementation	Apply per flow/provider/deployer role before public EU launch.

23. Definition of Done

- AI provider/domain ports, policy engine, budget controls, prompt registry and audit contracts are versioned and tested.
- At least one local/free adapter can be enabled without changing learner/admin feature contracts.
- Paid inference is provably blocked under default configuration.
- English and Czech reference fixtures pass AI explanation/hint bilingual evals and provider-offline fallback tests.
- AI-generated learner practice cannot silently inflate mastery.
- Admin AI cannot publish or execute privileged/destructive commands.
- All AI Auto actions are allowlisted, draft-scoped, diffable and audited.
- Outbound payload privacy allowlists and provider data-policy registry are active.
- No secrets, raw chain-of-thought or raw microphone bytes appear in AI logs/analytics.
- AI UX is labeled, accessible and has non-AI alternatives.
- Provider/model/prompt/policy versions appear in diagnostics and eval reports.
- Incident controls can disable provider or all AI without redeploy and without breaking core learning.
- Governing Formal/Technical/Curriculum/Reference documents are synchronized in the next controlled revisions.

Appendix A. Provider candidates verified 04.09.2026

This appendix is operational guidance, not a permanent vendor commitment. Free limits and data policies can change; every provider profile MUST be revalidated before enabling it in production.

Provider / class	Verified facts on 04.09.2026	Lingvessa default use
Gemini Developer API Free	Official pricing page offers a Free tier with limited model access and free input/output tokens. The page states Free-tier content is used to improve Google products.	Useful for development and low-risk synthetic/admin tasks; NOT default route for personal learner content without explicit policy approval.
Groq Free	Official docs publish Free Plan rate limits per model. Current data docs state inference customer data is not retained by default except limited feature/reliability cases and is not used for model training without permission.	Strong candidate for low-volume text AI after account/data-control review; exact model limits remain configurable.
OpenRouter Free	Official pricing currently lists 25+ free models and 50 requests/day on Free; free router can select among free providers/models. Underlying provider/model data policies vary.	Experimental/fallback for low-risk tasks only with provider-policy filtering; avoid blind random routing for sensitive learner data.
Ollama / local	Official API supports local model execution. No cloud inference fee is inherent to running a local model; capability depends on host hardware/model.	Preferred local adapter class for admin/dev and privacy-sensitive experiments where performance is sufficient.

Provider recommendation for start: implement adapters and policy first; then run a quality/privacy/quota spike. Do not hardcode a provider name into learner features or content schemas.

A.1. Provider selection criteria

- Ukrainian explanation quality.
- English en-GB/en-US and Czech grammar/morphology reliability.

- Structured JSON/tool support.
- Latency and rate limits.
- Data retention/training terms and regions.
- Free-tier sustainability and quota.
- Ability to disable retention where offered.
- Model/version stability and deprecation behavior.

Appendix B. AI feature-risk matrix

Feature	Risk	Evidence authority	Default mode
Explain differently	Low/medium	None by itself	AI on if approved provider
Adaptive hint	Medium	Assistance only	AI on selectively
Extra practice free-form	Medium	Practice-only	Feature flag
Extra practice deterministic template	Medium	Bounded mastery evidence	Feature flag
Writing feedback	Medium/high	Advisory until calibrated	Feature flag
AI conversation text	Medium	Practice; rubric only if validated	Feature flag
AI voice role-play	Medium/high	Speech evidence contract separate	Later
Admin draft generation	Medium	No publish authority	AI Assist
Admin QA suggestions	Low/medium	Issue/proposal only	AI Assist
Admin Auto draft metadata/jobs	Medium	Draft/job only	Off by default
Publish/delete/roles/billing	High	Human authority only	AI prohibited

Appendix C. Example contracts

C.1. AI request envelope

Illustrative only; canonical schema lives in repository.

```
{
  "request_id": "0198f0a0-1000-7abc-8def-100000000001",
  "capability": "learner_explain",
  "actor": {"type": "learner", "language_profile_id": "0198f0a0-1000-7abc-8def-100000000002"},
  "scope": {"language": "en-GB", "cefr": "A1", "knowledge_refs": ["0198f0a0-1000-7abc-8def-100000000003"]},
  "data_class": "pseudonymous_learning",
  "prompt_template_version": "explain_v1",
  "output_schema_code": "ai_explanation_v1",
  "budget_policy": {"free_only": true, "paid_allowed": false},
  "grounding": {"content_version_id": "0198f0a0-1000-7abc-8def-100000000004"}
}
```

C.2. AI result envelope

```
{
  "request_id": "0198f0a0-1000-7abc-8def-100000000001",
  "status": "ok | unavailable | policy_blocked | invalid_output",
  "result": {"...": "validated structured payload"},
  "grounding_refs": ["0198f0a0-1000-7abc-8def-100000000003", "0198f0a0-1000-7abc-8def-100000000004"],
  "versions": {"provider": "...", "model": "...", "prompt": "...", "policy": "..."},
  "usage": {"input_tokens": 0, "output_tokens": 0, "estimated_cost": 0},
  "correlation_id": "0198f0a0-1000-7abc-8def-100000000005"
}
```

C.3. Admin AI settings

```
{
  "ai_enabled": true,
  "learning_ai_enabled": true,
  "admin_ai_enabled": true,
  "admin_mode": "assist",
  "free_only": true,
  "paid_inference_enabled": false,
  "monthly_paid_budget": 0,
  "auto_allowlist": ["create_draft_issue", "create_media_job", "fill_draft_metadata"],
  "forbidden_actions": ["publish", "rollback", "delete_account", "change_role", "billing"]
}
```

Appendix D. Sources and change control

Internal governing sources:

- Lingvessa Formal Specification v1.7.
- Lingvessa Technical Specification v1.7.
- Lingvessa Curriculum Blueprint A1 v1.6.
- Lingvessa Reference Module Design v1.6.
- Lingvessa Technical Design v1.7 (09.09.2026), including two-level mastery and binary usable-speech UX with neutral speech_unavailable fallback.

External sources verified 04.09.2026:

https://ai.google.dev/gemini-api/docs/pricingGoogle AI for Developers - Gemini Developer API pricing: source
https://console.groq.com/docs/rate-limitsGroqDocs - Rate Limits: source
https://console.groq.com/docs/your-dataGroqDocs - Your Data in GroqCloud: source
https://openrouter.ai/pricingOpenRouter - Pricing: source
https://openrouter.ai/openrouter/free/providersOpenRouter - Free Models Router: source
https://docs.ollama.com/api/introductionOllama - API Introduction: source
https://digital-strategy.ec.europa.eu/en/library/guidelines-transparency-obligations-providers-and-deployers-ai-systemsEuropean Commission - Guidelines on transparency obligations under Article 50 of the AI Act: source
https://www.w3.org/TR/WCAG22/W3C - WCAG 2.2: source
https://owasp.org/www-project-application-security-verification-standard/OWASP - Application Security Verification Standard: source

D.1. Source interpretation

Provider pricing/free tiers and data policies are volatile. They are recorded as provider-registry metadata and operational candidates, not normative promises to the user. Before public launch or any provider change, Lingvessa rechecks current terms, data handling, regions, model availability, quotas and legal requirements.

D.2. Change control

Change	Required control
New learner AI capability	Product/pedagogical/privacy review + AI spec minor revision if authority/data flow changes.
New provider/model	Provider profile, privacy/terms review, adapter tests, bilingual evals and rollout flag.
Prompt/system policy	Versioned prompt + eval report; no overwrite of historical version.
AI Auto permission	Risk classification, RBAC/policy update, negative security test, audit and owner approval.
Paid inference enablement	Explicit owner approval, budget > 0, alerts, provider billing review and ADR.

Change	Required control
AI evidence affecting mastery	Calibration/validation study, deterministic/rubric contract and governing learning-spec revision.
Retention/data routing	Privacy inventory/legal review; user-facing policy update where needed.

APPROVAL RECORD: AI Integration Specification v1.4 is the owner-locked on 10.09.2026 for implementation P0; future revisions remain controlled and historical PDFs are not overwritten.

КІНЕЦЬ AI INTEGRATION SPECIFICATION v1.4

Controlled baseline synchronization - 09.09.2026

Normative baseline
This document is part of the synchronized Lingvessa owner-locked implementation baseline; P0 is authorized as of 10.09.2026. Earlier current-version references are superseded by the exact set below; historical "replaces" statements remain historical only.

Document	Version	Authority
Lingvessa_Formal_Specification	v1.7	Product scope, pedagogy, user-facing invariants
Lingvessa_Technical_Specification	v1.7	Normative technical requirements and interfaces
Lingvessa_Technical_Design	v1.7	Concrete implementation contracts, DDL/state/API design
Lingvessa_Curriculum_Blueprint_A1	v1.6	A1 pedagogical scope and module architecture
Lingvessa_Reference_Module_Design	v1.6	Executable E-A1-M01/C-A1-M01 fixtures
Lingvessa_AI_Integration_Specification	v1.4	AI-specific policy, data, routing, permissions
Lingvessa_Implementation_Plan	v1.4	Sole execution-order authority; no scope override

Decision	Topic	Fixed rule
BL-001	Baseline set	Formal 1.7 / Technical 1.7 / Design 1.7 / Curriculum A1 1.6 / Reference 1.6 / AI 1.4 / Plan 1.4
BL-002	Precedence	Domain-specific governing document controls; unresolved conflict -> owner decision + ADR/spec revision before code merge.
BL-003	Identity contract	id = UUID; stable_code = immutable human-readable semantic code; version_code = immutable revision label. Never put stable_code in *_id.
BL-004	Mastery contract	Exactly six global skills; canonical facet registry in this release. Reading is assessment-only domain, not seventh mastery skill.
BL-005	Language isolation	Composite DB FKs/checks/trigger guards enforce user + language_profile + language lineage for learner state and learner AI.
BL-006	Pronunciation states	speech_correct / speech_incorrect / speech_unavailable are canonical. Incorrect is non-blocking; Auto-advance occurs only after feedback dwell and never records success.

BL-007	Speech lifecycle	Asynchronous speech_job lifecycle with idempotent final attempt settlement and transient audio cleanup.
BL-008	AI API/data model	Canonical /api/v1/ai/explain|hint|practice|writing-feedback|conversations plus unified ai_* physical model.
BL-009	Monorepo	One /packages/provider-adapters package with ai/, speech/, tts/, image/, email/, storage/ sub-adapters; /api-client and /ui retained.
BL-010	Execution order	Implementation Plan v1.4 P0-P13 is the sole milestone numbering authority.
BL-011	Learner AI opt-in	Learner AI is OFF by default until explicit opt-in/consent. Effective enablement order is global -> capability -> provider/policy -> learner setting/consent -> quota.
BL-012	OpenAPI	Pin API description to OpenAPI Specification 3.1.2 for v1 baseline; upgrades require ADR/contract diff.

Precedence: Formal controls product/pedagogical scope; Curriculum controls A1 pedagogy; Technical Specification controls normative technical requirements; Technical Design controls concrete implementation detail; AI Specification controls AI-specific behavior; Reference Module Design is the executable fixture; Implementation Plan controls execution order only. Any unresolved conflict blocks implementation until an owner decision and controlled revision/ADR.

Canonical identity and version contract

Field	Type	Example	Rule
id	UUID	018f...	Opaque database/API identity.
stable_code	string	E-A1-M01-L01	Immutable human-readable semantic code; never stored in *_id.
version_code	string	1.4 / 2026-09-05.1	Immutable revision label for a versioned entity.
provider_external_id	string nullable	vendor-model-x	External/provider identifier only in adapter/config boundary.

API/JSON examples MUST use UUID-shaped values in *_id fields. Content manifests MAY use stable_code for authoring references, but the build/publish step resolves them to UUID + version_code and rejects unresolved codes.

Canonical AI API and persistence mapping

API	Purpose
POST /api/v1/ai/explain	Grounded explanation
POST /api/v1/ai/hint	Bounded hint
POST /api/v1/ai/practice	Extra practice
POST /api/v1/ai/writing-feedback	Writing feedback
POST /api/v1/ai/conversations	Create bounded conversation session
POST /api/v1/ai/conversations/{session_id}/messages	Conversation turn
DELETE /api/v1/ai/conversations/{session_id}	Delete/close learner AI session
Logical concept	**Physical entity/table**
AIProviderProfile	ai_provider_profiles
AIModelProfile	ai_model_profiles

AIPolicy / permissions / budget eligibility	ai_policies
AIPromptTemplate	ai_prompt_templates
AIRun (logical request record) / AIExecution / AIRun	ai_runs
AISession	ai_sessions
AIMessage	ai_messages
AIArtifact	ai_artifacts
AIUsage / budget consumption	ai_usage_buckets + audit_events
AIToolAction	ai_tool_actions

Learner AI is OFF by default. Effective enablement order: global AI flag -> capability flag -> approved provider/model/data policy -> learner opt-in and required consent -> quota/budget. Any failed check returns deterministic fallback; no paid fallback is implicit.

Canonical execution authority and milestone map

Milestone	Authority
P0	Governance freeze, exact baseline, decision log, full MUST trace matrix
P1	Monorepo/bootstrap/CI/config
P2	Data, UUID/stable_code/version_code, language isolation, publication
P3	Lesson Engine + attempts + deterministic state
P4	Two-level Mastery + Review + mistakes + collections
P5	Media/audio/speech + async speech jobs
P6	English E-A1-M01
P7	Czech C-A1-M01 + bilingual isolation proof
P8	Admin/CMS + publication + Admin AI Assist foundation
P9	Optional learner AI + AI Auto allowlist
P10	Onboarding/Home/Explore/Course/Progress/Profile/Settings/ notifications + persistence
P11	Reference hardening + 32 scenarios + technical gates; freeze reference baseline
P12	A1 M02-M12 controlled rollout + pilot/calibration + optional A1 Level Test
P13	Full-system hardening/launch gate: security, privacy, performance, restore/DR, release provenance, legal/product readiness

Only Lingvessa_Implementation_Plan v1.4 uses P0-P13 numbering. Any older P-number labels in other specifications are descriptive history and MUST be mapped to these milestones rather than treated as execution authority.

Canonical external references

- OpenAPI Specification 3.1.2
- CEFR Companion Volume
- Council of Europe - CEFR
- English Profile
- Čeština pro cizince

OpenAPI here means the OpenAPI Specification for describing HTTP APIs; it is unrelated to OpenAI as an AI provider. OpenAI remains only a possible future provider behind the provider-agnostic AI adapter and is not required by the v1 baseline.

Approval record

Поле	Значення

Baseline status	OWNER-LOCKED IMPLEMENTATION BASELINE / P0 AUTHORIZED
Approval date	10.09.2026
Approver	Product owner
Candidate version	v1.4
Accepted exceptions	None
Execution authority	Lingvessa_Implementation_Plan v1.4; P0 must be closed before P1.

AI example identity rule: request_id, correlation_id, actor_id, language_profile_id, provider_id, model_id and other *_id fields are UUIDs. Semantic identifiers such as output schema names use *_code (for example output_schema_code).